alodi

Proof document

The numbers behind Alodi

How one area of 20 couriers and 4,000 orders a day loses ₫473M a month to failed deliveries, and how Alodi cuts 41% of it. Every input, formula and source, in one place.

₫473M	₫215M	₫21M	₫194M
LOST A MONTH TODAY MODELLED	RECOVERED A MONTH PROJECTED	RUNNING COST A MONTH ESTIMATE	NET SAVING, 41% PROJECTED

Labels · **Sourced** public document · **Vendor** company claim · **Listed price** shop or price list · **Estimate** team assumption · **Projected** not yet measured

01 · INPUTS

Twelve inputs. Each one typed.

Every later number is arithmetic on this table. Change an input in cost_model.py and the whole document regenerates.

INPUT	VALUE	TYPE	SOURCE
Orders a day, one area	4,000	ESTIMATE	Team sizing: 20 couriers × 200 stops
First-attempt failure rate	8%	SOURCED	VECOM 8–12% nationwide, low end (Nguoi Lao Dong, 19 Jul 2025)
Rescued today (baseline)	25%	VENDOR	Shipsy: industry rescue rate 20–30%
Return fee per failed order	₫40,000	SOURCED	Shopee Return & Refund Policy §7.1 (seller pays on failed orders); cap ₫40,000 since 1 Apr 2025, GHN
Extra attempts per failed order	1.5	ESTIMATE	Inside Shopee's 2–3 contact attempts (Help Center 79569)
Courier pay per attempt	₫6,000–8,000	SOURCED	GHN blog, courier earnings per order
Courier income a month	₫8–15M	SOURCED	Glints Vietnam, full-time courier income
Courier hours a month	234	ESTIMATE	26 days × 9 hours
Door time lost per failure	5 min	ESTIMATE	Two calls plus waiting at the door
CS calls per failure × minutes	2 × 4 min	ESTIMATE	Team assumption
CS cost per hour	₫35,000	ESTIMATE	Team assumption
Days a month · exchange rate	30 · ₫26,000/$	ESTIMATE	Prices checked 30 Oct 2026

> Not counted, so the loss is conservative: the seller's lost sale, packaging, ad spend, damaged goods, and buyers losing COD after three failed orders in 90 days (Shopee Help Center 79196).

02 · WHAT ONE AREA LOSES

₫446.6M–₫499.3M a month. Midpoint ₫472.9M.

4,000 orders × 8% = **320 failed a day**. Shipsy's 25% baseline rescues 80; **240 go back** to the seller.

Return fees paid by sellers	320 failed × 75% not rescued = 240/day × ₫40,000 × 30	₫288.0M
Courier pay for redelivery	320 × 1.5 attempts × ₫6,000–8,000 × 30	₫86.4M–₫115.2M
Courier time at unanswered doors	320 × 5 min × (₫8–15M ÷ 234 h) × 30	₫27.4M–₫51.3M
CS calling buyers back	320 × 2 calls × 4 min × ₫35,000/h × 30	₫44.8M
Total a month	sum of the four lines · midpoint = (low + high) ÷ 2	**₫446.6M–₫499.3M**

A year

≈ ₫5.7B

Midpoint × 12.

At VECOM's 12% upper rate

≈ ₫670M–₫749M

Every line scales by 1.5×. The headline keeps the low end.

03 · WHAT ALODI COSTS

₫20.9M a month to run. 4.4% of the loss.

Local-first: one workstation per area runs OpenAI's open-weight gpt-oss-20b, PhoWhisper and Piper. The OpenAI API is only a masked fallback for ~10% of requests.

MONTHLY RUNNING COST	VND
OpenAI API fallback (10% of LLM calls, gpt-5-mini)	₫0.23M
OpenAI TTS for dynamic phrases (gpt-4o-mini-tts)	₫0.61M
Local models: gpt-oss-20b, PhoWhisper, Piper TTS	₫0.00M
Viettel Mobile SIP trunk (calls + line)	₫3.19M
Zalo ZNS (risky only)	₫11.25M
Electricity (≈ 272 kWh)	₫0.82M
Business fibre with static IP	₫0.80M
Off-site encrypted backup (Amazon S3, ~10 GB)	₫0.05M
Remote ops 0.1 FTE (shared across hubs)	₫3.60M
Hardware upkeep reserve (5% of hardware a year)	₫0.36M
Total	**₫20.9M**

ONE-TIME HARDWARE PER AREA (NOT IN THE MONTHLY FIGURES)	VND
2 × NVIDIA RTX 5060 Ti 16 GB (Phong Vu, ₫16.4–21.0M each)	₫32.8M–₫42.0M
Workstation: 12–16-core CPU, 64 GB RAM, 2 × 2 TB NVMe (RAID 1), 1000 W PSU, case	₫35.0M–₫45.0M
UPS 1.5–2 kVA (online)	₫5.0M–₫8.0M
4G/5G failover router	₫2.0M–₫3.0M
Total	**₫74.8M–₫98.0M**

> For comparison, the AWS-managed design runs at ₫68.4M a month, so local-first is about 69% cheaper.

04 · WHAT ALODI SAVES

₫193.8M net a month = 41.0% of the loss.

The headline assumes today's 25% rescue rate rises to 65%, the top of Shipsy's published best-in-class range (vendor, 50–65%). Base and conservative cases are shown next to it.

SAVING ITEM · FORMULA	CONSERVATIVE 15 pts · 1.50 · 40%	BASE 20 pts · 1.35 · 60%	BEST-IN-CLASS 40 pts · 1.20 · 65%
Return fees avoided 320 × extra rescue points × ₫40,000 × 30	₫57.6M	₫76.8M	**₫153.6M**
Redelivery attempts avoided 320 × (1.5 − attempts after) × ₫7,000 × 30	₫0.0M	₫10.1M	**₫20.2M**
Courier door time saved attempts saved × 5 min × ₫49k/h × 30	₫0.0M	₫5.9M	**₫11.8M**
CS calls handled by the agent 320 × 2 × 4 min × ₫35,000/h × agent share × 30	₫17.9M	₫26.9M	**₫29.1M**
Gross saving a month	**₫75.5M**	**₫119.7M**	**₫214.7M**
− Running cost	₫20.9M	₫20.9M	₫20.9M
Net saving a month	**₫54.6M**	**₫98.8M**	**₫193.8M**
Net ÷ monthly loss (₫472.9M)	11.5%	20.9%	41.0%
Net a year	₫0.66B	₫1.19B	₫2.33B
Hardware pays back in	7 weeks	4 weeks	2 weeks

41% = ₫193.8M ÷ ₫472.9M · ₫193.8M = ₫214.7M − ₫20.9M

Columns: extra rescue points · redelivery attempts per failed order after Alodi · share of CS calls the agent handles. Everything on this page is projected until a pilot measures Alodi's own rescue rate.

05 · SURVEY

12 couriers, 100 buyers.

Couriers from SPX Express (4), GHN (2), J&T Express (2), BEST Express (2), GHTK (1), Viettel Post (1). Buyers by segment: busy office worker 36, student 21, homemaker / retired 15, shift worker 15, self-employed / online seller 13. Instrument in Vietnamese and English (prep/06-survey/codebook.csv, 45 coded columns).

Data status · MOCK 112. The response files in the repository are still the generated placeholders, so the values below are illustrative and are not quoted as findings anywhere. Replace couriers.csv and customers.csv with the interview export (data_status = REAL) and rebuild this document.

WHO	KEY QUESTIONS	ANSWERS
cust	In the last 3 months, how often did a courier call while you were busy (meeting, driving, working)? 3 tháng qua, shipper gọi lúc bạn đang bận (họp, chạy xe, làm việc) bao nhiêu lần?	1 Never … 5 Very often
cust	How often did you miss the courier's call? Bạn lỡ cuộc gọi của shipper thường xuyên không?	1 Never … 5 Very often
cust	How often did you call the courier back and could not reach them? Bạn gọi lại shipper nhưng không liên lạc được bao nhiêu lần?	1 Never … 5 Very often
cust	How often did you have to remind the courier of a time or instruction you had already given? Bạn phải nhắc lại shipper giờ hẹn/lời dặn đã nói trước đó bao nhiêu lần?	1 Never … 5 Very often
cust	I would like to choose a delivery time window Tôi muốn được chọn khung giờ giao hàng	1 Strongly disagree … 5 Strongly agree
cust	I am comfortable receiving an automated call from the carrier Tôi thoải mái nhận cuộc gọi tự động từ đơn vị vận chuyển	1 … 5
cust	If the carrier messaged you to pick a time and remembered your instructions, would you use it? Nếu đơn vị vận chuyển nhắn hỏi giờ và nhớ lời dặn của bạn, bạn có dùng không?	Yes | Maybe | No
cour	Calls you make to buyers per day Số cuộc gọi cho khách mỗi ngày	≥ 0
cour	Share of your calls that buyers do not answer (%) % cuộc gọi khách không nghe	0–100
cour	If a system asked buyers for a time and gave you a card with the time, drop-off and backup contact, saving you time and keeping everyone on the same page, would you like it? Nếu có hệ thống hỏi giờ khách trước và đưa anh/chị thẻ ghi giờ, nơi để hàng, người nhận thay — tiết kiệm thời gian và thống nhất hơn — anh/chị có thích không?	Yes | Not sure | No
cour	Minutes you think it would save per day Ước tính tiết kiệm bao nhiêu phút mỗi ngày	≥ 0

05 · SURVEY · HEADLINE VALUES

What the survey measures.

Shares and means computed from the response files by generate_mock_survey.py / the EDA notebook. Shown as placeholders until the interview export replaces them.

WHO	HEADLINE	PLACEHOLDER	N
customer	Courier called while busy, often or very often	56.0%	100
customer	Missed the courier's call, often or very often	39.0%	100
customer	Called back and could not reach the courier, sometimes or more	66.0%	100
customer	Had to remind the courier, sometimes or more	81.0%	100
customer	Instruction not followed, sometimes or more	59.0%	100
customer	Had an order returned or cancelled in the last 3 months	31.0%	100
customer	Want to choose a delivery window (agree or strongly agree)	86.0%	100
customer	Comfortable with an automated call (agree or strongly agree)	58.0%	100
customer	Would use an Alodi-like service: Yes	65.0%	100
customer	Would use an Alodi-like service: Maybe	34.0%	100
customer	Would use an Alodi-like service: No	1.0%	100
courier	Would like an Alodi-like system: Yes	83.3%	12
courier	Would like an Alodi-like system: Not sure	16.7%	12
courier	Would like an Alodi-like system: No	0.0%	12
courier	Buyers call while riding, often or very often	83.3%	12
courier	Keep buyers' instructions only in memory	25.0%	12
courier	Mean calls to buyers per day	107.8	12
courier	Mean unanswered call share (%)	26.6	12
courier	Mean minutes per day on calls	78.9	12
courier	Mean expected minutes saved per day (Yes answers)	46.0	12
customer	Mean pain index (1–5)	3.2	100

06 · THE SIMULATED DAY BEHIND THE DASHBOARD

One fictional hub, fixed seed 31102026.

3,993 orders	**3,447** delivered first time	**350** failed: A 209 · B 91 · refused 50
231 rescued with a context card	**116** to customer service	**187** AI calls, 0 out of hours

Rates are team assumptions for testing the product, not measurements. The landing page's problem figures do not use this simulation; they use the cost model above.

07 · SOURCES

- Shopee Return & Refund Policy §7.1–7.2 (checked 31 Oct 2026) — help.shopee.vn/portal/4/article/77251
- GHN, who pays for Shopee returns — ₫40,000 cap per order since 1 Apr 2025 (₫20,000 express) — ghn.vn/blogs/tip-ban-hang/tra-hang-shopee-ai-chiu-phi
- VECOM and Tran Anh Tung (UEF), Nguoi Lao Dong, 19 Jul 2025 — refusal rates 8–12%, 25–30% in sale season
- Shopee Help Center, articles 79559 (no delivery time choice), 79569 (2–3 contacts, 5 days), 79196 (COD lock)
- GHN blog — courier earnings ₫6,000–8,000 per order
- Glints Vietnam — courier income ₫8–15M a month
- Shipsy, What is NDR management — industry rescue 20–30%, best-in-class 50–65% (vendor)
- Phong Vu listings — RTX 5060 Ti 16 GB ₫16.4–21.0M; Viettel Mobile SIP reseller price list; oazns.vn / worldfone ZNS price tables; EVN business tariff
- OpenAI pricing for gpt-5-mini and gpt-4o-mini-tts (third-party price lists, checked 30 Oct 2026)
- Model: prep/02-technical/cost-model/cost_model.py · PRD §18.8 · this document: build_proof.py